

Zaiqiao Meng

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Research Interests

- **Natural Language Understanding and Reasoning:** combining natural language models with knowledge graphs and reasoning techniques to enhance various tasks such as document classification, named entity recognition, entity linking, question answering.
- **Biomedical Informatics:** combining natural language processing, knowledge graph, epidemiology, biomedical informatics and bioethics to address complex biomedical task.
- **Machine Learning on Relational Data:** building practical machine learning and representation learning models to handling complex, dynamic relational data.
- **Recommender Systems:** modelling various side information, such as user social, item textual and their dynamics in recommenders systems

Education

- **Sun Yat-sen University** **Guangzhou, China**
Ph.D. in Computer Science, Adviser: Prof. Hong Shen *December 2018*
- **Guangdong University of Technology** **Guangzhou, China**
M.Eng. in Computer Science, *July 2014*
- **Jiangxi University of Science and Technology** **Ganzhou, China**
B.Eng. in Computer Science, *July 2010*

Work Experience

- **University of Glasgow** **Glasgow, UK**
Lecturer in Health Technologies, *January 2022 - Present*
- **University of Cambridge** **Cambridge, UK**
Research Associate, Adviser: Prof. Nigel Collier *July 2020 - January 2022*
Work on the EPI-AI project, an ESRC project aiming at Automated Understanding and Alerting of Disease Outbreaks from Global News Media.
 - Work on infusing biomedical knowledges (e.g. UMLS) into pretrained language models (e.g., BioBERT).
 - Implement various NLP models for the EPI-AI project, including document classification, entity linking, NER and question answering.
 - Co-supervise MPhil and Ph.D. students.
- **University of Glasgow** **Glasgow, UK**
Research Assistant, Adviser: Prof. Iadh Ounis *March 2019 - July 2020*
Work on the bigdatastack project, a project funded under the European Commission Horizon 2020 Work Programme.
 - Work on the research task of Grocery Recommender Systems, proposed new models that consider various involving side information and improve the recommendation effectiveness.
 - Initialise and led the Beta-Recsys project, an open source project for Building, Evaluating and Tuning Automated Recommender Systems.
 - Co-supervise master and Ph.D students.

- **King Abdullah University of Science and Technology** **Thuwal, Saudi Arabia**
- **Visiting Ph.D, Adviser: Prof. Xiangliang Zhang** **March 2018 - August 2018**
 - Worked on the research task of (dynamic) attributed network embedding, proposed two effective embedding models that can benefit various applications such as link prediction and node classification.

Teaching Experience

- **Guest Lecturer:** Recommender Systems, University of Glasgow *Aug. 2021*
- **Guest Lecturer:** Recommender Systems, University of Glasgow *Mar. 2021*
- **Guest Lecturer, Lab Tutor:** Recommender Systems, University of Glasgow *Mar. 2020*

Selected Talks

1. **University of Glasgow** **Virtual**
Biomedical Knowledge-Enhanced Language Modelling *15-Feb-2021*
2. **SIGIR2020 - TOIS invited talk** **Virtual**
Jointly Learning Representations of Nodes and Attributes for Attributed Networks *27-July-2020*
3. **University of Glasgow** **Glasgow, UK**
Variational Bayesian Context-aware Representation for Grocery Recommendation *24-Oct-2019*

Publications

- **Google Scholar Citations: 369**
- **H-index: 11**

* indicates equal contributions; † indicates corresponding author(s)

Publication information updated on 21 Dec. 2021

Conference Papers

1. Jinyuan Fang, Qiang Zhang, **Zaiqiao Meng**, and Shangsong Liang. Structure-Aware Random Fourier Kernel for Graphs. *NeurIPS* 2021.
2. Qiang Zhang, Jinyuan Fang, **Zaiqiao Meng**, Shangsong Liang, and Emine Yilmaz. Variational Continual Bayesian Meta-Learning. *NeurIPS* 2021.
3. Yixuan Su, **Zaiqiao Meng**, Simon Baker and Nigel Collier, Few-Shot Table-to-Text Generation with Prototype Memory. *EMNLP* 2021
4. **Zaiqiao Meng**, Fangyu Liu, Thomas Clark, Ehsan Shareghi and Nigel Collier. Mixture-of-Partitions: Infusing Large Biomedical Knowledge Graphs into BERT, *EMNLP* 2021
5. Jinyuan Fang, Shangsong Liang, **Zaiqiao Meng** and Qiang Zhang. Gaussian Process with Graph Convolutional Kernel for Relational Learning. *SIGKDD* 2021. (Core A*, CCF A)
6. Siyuan Liao, Shangsong Liang, **Zaiqiao Meng** and Qiang Zhang. Learning Dynamic Embeddings for Temporal Knowledge Graphs. *WSDM* 2021. (Core A*, CCF B)
7. Fangyu Liu, Ehsan Shareghi, **Zaiqiao Meng**, Marco Basaldella and Nigel Collier. Self-alignment Pre-training for Biomedical Entity Representations. *NAACL* 2021. (Core A, CCF C)
8. Jinpeng Wang, Bin Chen, Qiang Zhang, **Zaiqiao Meng**, Shangsong Liang and Shutao Xia. Weakly Supervised Deep Hyperspherical Quantization for Image Retrieval. *AAAI* 2021. (Core A*, CCF A)
9. **Zaiqiao Meng**, Richard McCreadie, Craig Macdonald, Iadh Ounis, and others, BETA-Rec: Build, Evaluate and Tune Automated Recommender Systems. *ACM RecSys* 2020. (Core A)
10. **Zaiqiao Meng**, Richard McCreadie, Craig Macdonald, Iadh Ounis, Exploring Data Splitting Strategies for the Evaluation of Recommendation Models. *ACM RecSys* 2020. (Core A)
11. Siwei Liu, Iadh Ounis, Craig Macdonald, **Zaiqiao Meng**. A Heterogeneous Graph Neural Model for Cold-start Recommendation. *SIGIR* 2020. (Core A*, CCF A)

12. **Zaiqiao Meng**, Jinyuan Fang, Shangsong Liang, Teng Xiao. Semi-supervisedly Co-embedding Attributed Networks. *NeurIPS* 2019. (Core A*, CCF A)
13. **Zaiqiao Meng**, Shangsong Liang, Xiangliang Zhang. Co-Embedding Attributed Networks. *WSDM* 2019. (Core A*, CCF B)
14. Teng Xiao, Jiaxin Ren, **Zaiqiao Meng**, Huan Sun, Shangsong Liang. Dynamic Bayesian Metric Learning for Personalized Product Search. *CIKM* 2019. (Core A, CCF B)
15. Yupeng Luo, Shangsong Liang, **Zaiqiao Meng**. Constrained Co-embedding for User Profile in Community Question Answering. *CIKM* 2019. (Core A, CCF B)
16. Teng Xiao, Shangsong Liang, **Zaiqiao Meng**. Dynamic Collaborative Recurrent Learning. *CIKM* 2019. (Core A, CCF B)
17. Teng Xiao, Shangsong Liang, Wenzhou Shen, **Zaiqiao Meng**. Bayesian Deep Collaborative Matrix Factorization. *AAAI* 2019. (Core A*, CCF A)
18. Teng Xiao, Shangsong Liang, **Zaiqiao Meng**. Hierarchical Neural Variational Model for Personalized Sequential Recommendation. *WWW* 2019, Short paper. (Core A*, CCF A)
19. **Zaiqiao Meng**, Hong Shen. NMFDIV: A Nonnegative Matrix Factorization Approach for Search Result Diversification on Attributed Networks. *PDCAT*. 2017. (Core B)

Journal Articles.....

1. Shangsong Liang, Shaowei Tang, **Zaiqiao Meng**[†], Qiang Zhang . Cross-Temporal Snapshot Alignment for Dynamic Networks. *Submitted to IEEE Transactions on Knowledge and Data Engineering*. (CCF A)
2. Jinyuan Fang, Shangsong Liang, **Zaiqiao Meng**, Maarten de Rijke. Hyperspherical Variational Co-embedding for Attributed Networks. *ACM Transactions on Information Systems*, 2021. (CCF A, IF. 4.676).
3. Zuo Ouyang, **Zaiqiao Meng**, Qiang Zhang, Shangsong Liang. A Normalizing Flow-based Co-embedding Model for Attributed Networks. *IEEE Transactions on Knowledge Discovery from Data*, 2021. (CCF B, IF. 3.089).
4. **Zaiqiao Meng**, Richard McCreadie, Craig Macdonald, Iadh Ounis. Variational Bayesian Representation Learning for Grocery Recommendation. *Information Retrieval Journal*, 2021. (IF. 2.744)
5. Yaoxin Pan, Shangsong Liang, Jiaxin Ren, **Zaiqiao Meng**, Qiang Zhang. Personalized, Sequential, Attentive, Metric-Aware Product Search. *ACM Transactions on Information Systems*, 2021. (CCF A, IF. 4.676).
6. Shangsong Liang, Yupeng Luo, **Zaiqiao Meng**[†]. Profiling Users for Question Answering Communities via Flow-based Constrained Co-embedding Model. *ACM Transactions on Information Systems*, 2021. (CCF A, IF. 4.676)
7. Huimin Huang, **Zaiqiao Meng**[†], Hong Shen. Competitive and complementary influence maximization in social network: A follower's perspective. *Knowledge-Based Systems*. 2021. (CCF B, IF. 5.921)
8. **Zaiqiao Meng**, Shangsong Liang, Xiangliang Zhang, Richard McCreadie, Iadh Ounis. Jointly Learning Representations of Nodes and Attributes for Attributed Networks. *ACM Transactions on Information Systems*. 2020. (CCF A, IF. 4.676)
9. Huimin Huang, Hong Shen, **Zaiqiao Meng**. Community-based influence maximization in attributed networks. *Applied Intelligence*. 2020. (CCF C, IF. 3.325)
10. Huimin Huang, **Zaiqiao Meng**[†], Shangsong Liang. Recurrent neural variational model for follower-based influence maximization. *Information Sciences* 2020. (CCF B, IF. 5.910)
11. **Zaiqiao Meng**, Hong Shen. Fast top-k similarity search in large dynamic attributed networks. *Information Processing & Management*. 2019. (CCF B, IF. 4.787)
12. Huimin Huang, Hong Shen, **Zaiqiao Meng**. Item diversified recommendation based on influence

- diffusion. *Information Processing & Management*, 2019. (CCF B, IF. 4.787)
13. **Zaiqiao Meng**, Hong Shen. Scalable Aspects Learning for Intent-aware Diversified Search on Social Networks. *IEEE Access*. 2018. (IF. 3.745)
 14. **Zaiqiao Meng**, Hong Shen, Huimin Huang, Wei Liu, Jing Wang, Arun Kumar Sangaiah. Search Result Diversification on Attributed Networks via Nonnegative Matrix Factorization. *Information Processing and Management*. 2018. (CCF B, IF. 4.787)
 15. **Zaiqiao Meng**, Hong Shen. Dissimilarity-Constrained Node Attribute Coverage Diversification for Novelty-Enhanced Top-k Search in Large Attributed Networks. *Knowledge-Based Systems*. 2018. (CCF B, IF. 5.921)

Technique Reports

1. **Zaiqiao Meng**, Richard McCreadie, Craig Macdonald, Iadh Ounis. Variational Bayesian Context-aware Representation for Grocery Recommendation. CARS2.0 Workshop at RecSys. 2019.

Under Review/Preprint

1. **Zaiqiao Meng**, Fangyu Liu, Ehsan Shareghi, Yixuan Su, Charlotte Collins, and Nigel Collier. Rewire-then-Probe: A Contrastive Recipe for Probing Biomedical Knowledge of Pre-trained Language Models. arXiv preprint arXiv:2110.08173 (2021).
2. Yixuan Su, Fangyu Liu, **Zaiqiao Meng**, Lei Shu, Ehsan Shareghi, and Nigel Collier. TaCL: Improving BERT Pre-training with Token-aware Contrastive Learning. arXiv preprint arXiv:2111.04198 (2021).
3. **Zaiqiao Meng**, Richard McCreadie, Craig Macdonald, Iadh Ounis. Temporal Variational Bayesian Representation Learning for Grocery Recommendation. *Under review*.
4. **Zaiqiao Meng***, Siwei Liu*, Craig Macdonald, Iadh Ounis. Graph Neural Pre-training for Recommendation with Side Information. *Under review*.
5. Shaowei Tang*, **Zaiqiao Meng***, Shangsong Liang. Dynamic Co-embedding Model for Temporal Attributed Networks. *Under review*.
6. Jiaxin Ren, Teng Xiao, **Zaiqiao Meng**, Shangsong Liang. Dynamic Product Search in Metric Space. *Under review*.
7. Yu Wang, Xin Xin, **Zaiqiao Meng**, Xiangnan He, Joemon Jose, Fuli Feng. Probabilistic and Variational Recommendation Denoising. *Under review*.
8. Qiang Zhang, Shangsong Liang, Hongbin Huang, **Zaiqiao Meng**, Emine Yilmaz. Learning to Detect Few-Shot-Few-Clue Misinformation. *Under review*.

Software

- Beta-RecSys**

 - *Build, Evaluate and Tune Automated Recommender Systems*
 - Initialize the project and build the core framework.
 - Manage a team with 11 members on maintaining the project.
 - Implement more than 15 recommendation models and integrate more than 20 datasets .
 - Contain various features, including hyper-parameter tuning, unified PyTorch core engine and unified training, validation, tuning and testing pipeline.

Leader, Main contributor
Github Open Source Project
- SCAN**

 - *Semi-supervisedly Co-embedding Attributed Networks*
 - Semi-supervisedly learn low-dimensional representations of both attributes and nodes in the same semantic space.
 - Benefit many applications such as node classification, link prediction and attribute inference.

Leader, Main contributor
Network Embedding Model
- CAN**

 - *Co-embedding Attributed Networks*

Leader, Main contributor
Network Embedding Model

- Learn low-dimensional representations of both attributes and nodes in the same semantic space.
- Benefit not only node-oriented network problems (e.g., node classification and link prediction), but also attribute inference problems (e.g., predicting the value of attributes of nodes).

Professional Activities

Conference Program Committee.....

○ ICML	2021
○ NeurIPS	2020, 2021
○ ICLR	2021,2022
○ SIGIR	2021
○ SIGKDD	2021
○ TheWebConf	2021,2022
○ SDM	2022
○ AAAI	2020, 2021,2022
○ IJCAI	2020
○ ECIR	2020, 2021, 2022
○ EMNLP	2021
○ RecSys	2020, 2021

Invited Journal Reviewer.....

○ ACM Transactions on Information Systems	2020,2021
○ IEEE Transactions on Neural Networks and Learning Systems	2020,2021
○ IEEE Transactions on Knowledge and Data Engineering	2020,2021
○ ACM Computing Surveys	2020
○ IEEE Transactions on Cybernetics	2020
○ Information Sciences	2019
○ IEEE Transactions on Fuzzy Systems	2019
○ IEEE Access	2019,2020
○ International Journal of Machine Learning and Cybernetics	2017
○ Concurrency and Computation: Practice and Experience	2017

External Reviewer.....

○ WSDM	2017, 2019
○ CIKM	2017, 2018
○ SIGIR	2017, 2018
○ EMNLP	2019
○ IJCAI	2018
○ WWW	2018
○ ACM Transactions on the Web	2017